Printed Pages:02 **Sub Code:KAS-103T**

Paper Id: 233426 **Roll No.** | | | | | | | | | | |

B. TECH.
(SEM I) THEORY EXAMINATION 2022-23
ENGINEERING MATHEMATICS-I

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. If $A=\begin{bmatrix}\alpha & \beta\\ \gamma & \delta\end{bmatrix}$ and $\alpha\delta-\beta\gamma=1$, then A^{-1}

b. If the Eigen values of the matrix A are $1,2,3$ then find the Eigen values of $(A.I^2)$

c. State Lagrange's mean value theorem.

d. What is the eighth derivative of $y=3x^8$?

e. If $u=e^x\sin y$ and $v=e^x\cos y$, Evaluate $\dfrac{\partial(u,v)}{\partial(x,y)}$.

f. What is the maximum value of function $f(x,y)=1-x^2-y^2$?

g. What does $\int_0^1\int_0^{1-x} dy\,dx$ represent?

h. Evaluate $\int_0^1\int_0^x xy\,dy\,dx$

i. Find the $grad\,\phi$ when ϕ is given by $\phi=3x^2y-y^3z^2$ at the point $(1,-2,-1)$.

j. Find the directional derivatives of $\dfrac{1}{r}$ in the direction of $\vec{r}$ where $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$.

SECTION B

2. Attempt any *three* of the following: **10x3=30**

a. Find the value of P for which the matrix $A=\begin{bmatrix}3 & P & P\\ P & 3 & P\\ P & P & 3\end{bmatrix}$ is of rank 1.

b. If $y=(\sinh^{-1}x)^2$ then find $(y_n)_0$.

c. Prove by Maclaurin's Theorem that $e^{\sin x}=1+x+\dfrac{x^2}{1.2}-\dfrac{3.x^4}{1.2.3.4}+\ldots$

d. Evaluate by changing the order of integration $\int_0^1\int_{2y}^{2} e^{x^2}\,dxdy$

e. If $\vec{F}=3y\,\hat{i}-xz\,\hat{j}+yz^2\,\hat{k}$ and S is the surface of the paraboloid $2z=x^2+y^2$ bounded by $z=2$, show by using Stoke's theorem $\iint_S(\vec{\nabla}\times\vec{F}).d\vec{S}=-20\pi$

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SECTION C

3. Attempt any *one* part of the following: 10x1=10

a. Investigate, for what values of λ and μ do the system of equations $x+y+z=6; x+2y+3z=10; x+2y+\lambda z=\mu$ have (i) no solution (ii) unique solution (iii) infinite solution

b. State Cayley-Hamilton Theorem and verify it for the matrix;

$$A=\begin{matrix}1 & 1 & 2\\ 9 & 2 & 0.\\ 5 & 0 & 3\end{matrix}$$ Also find A^{-1}.

4. Attempt any *one* part of the following: 10x1=10

a. If $x^x y^y z^z = c$, show that at $x=y=z$, $\dfrac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$

b. Trace the curve: $x^3 + y^3 = 3axy$

5. Attempt any *one* part of the following: 10x1=10

a. Find the possible % error in computing the parallel resistance r of three resistances r_1, r_2, r_3 from the formula $\dfrac{1}{r} = \dfrac{1}{r_1} + \dfrac{1}{r_2} + \dfrac{1}{r_3}$ if r_1, r_2, r_3 are each in error by $+1.2\%$.

b. Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.

6. Attempt any *one* part of the following: 10x1=10

a. Evaluate $\iint_R (x-y)^4 e^{x+y}\,dx\,dy$ where R is the square with vertices at $(1,0),(2,1),(1,2)$ and $(0,1)$.

b. Evaluate: $\int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dy\,dx\,dz$

7. Attempt any *one* part of the following: 10x1=10

a. If $\vec{F} = (\vec{a}.\vec{r})\vec{r}$, where $\vec{a}$ is a constant vector, find $curl\vec{F}$ and prove that it is perpendicular to $\vec{a}$.

b. Evaluate $\iint_S (a^2x^2 + b^2y^2 + c^2z^2)^{-1/2}\,dS$ where S is the surface of the ellipsoid $ax^2 + by^2 + cz^2 = 1$.

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